

Experiment No: 12

Title

Write a Python Program for Tic Tac Toe Game

Aim

To develop a Python program to implement the **Tic Tac Toe** game for two players and determine the winner or draw based on the game rules.

Objective

To implement a two-player Tic Tac Toe game using Python and verify the winning conditions after each move.

Algorithm

Step 1: Start the program.

Step 2: Initialize a 3×3 game board with empty spaces.

Step 3: Display the board.

Step 4: Allow Player X and Player O to enter their moves alternately.

Step 5: Check whether the entered position is empty.

Step 6: After every move, check for a winning condition.

Step 7: If a player wins, display the winner.

Step 8: If all positions are filled without a winner, declare a draw.

Step 9: Stop the program.

Program

```
# Tic Tac Toe Game
```

```
board = [' ' for _ in range(9)]
```

```
def display_board():
```

```
    print()
```

```
print(board[0], "|", board[1], "|", board[2])
print("--+---+--")
print(board[3], "|", board[4], "|", board[5])
print("--+---+--")
print(board[6], "|", board[7], "|", board[8])
print()
```

```
def check_win(player):
```

```
    win = [
        (0,1,2), (3,4,5), (6,7,8),
        (0,3,6), (1,4,7), (2,5,8),
        (0,4,8), (2,4,6)
    ]
```

```
    for x, y, z in win:
```

```
        if board[x] == board[y] == board[z] == player:
            return True
```

```
    return False
```

```
def board_full():
```

```
    return ' ' not in board
```

```
player = 'X'
```

```
while True:
```

```
display_board()
```

```
position = int(input(f'Player {player}, Enter Position (1-9): ')) - 1
```

```
if position < 0 or position > 8:
```

```
    print("Invalid Position")
```

```
    continue
```

```
if board[position] != ' ':
```

```
    print("Position Already Filled")
```

```
    continue
```

```
board[position] = player
```

```
if check_win(player):
```

```
    display_board()
```

```
    print("Player", player, "Wins!")
```

```
    break
```

```
if board_full():
```

```
    display_board()
```

```
    print("Match Draw")
```

```
    break
```

```
if player == 'X':
```

```
    player = 'O'
```

```
else:
```

```
    player = 'X'
```

Sample Input

Player X, Enter Position (1-9): 1

Player O, Enter Position (1-9): 5

Player X, Enter Position (1-9): 2

Player O, Enter Position (1-9): 8

Player X, Enter Position (1-9): 3

Sample Output

```
|
|
|
+---+
|
|
+---+
|
|
|

Player X, Enter Position (1-9): 1

X | | 
+---+
|
|
+---+
|
|
|

Player O, Enter Position (1-9): 5

X | | 
+---+
| O | 
+---+
|
|
|

Player X, Enter Position (1-9): 2

X | X | 
+---+
| O | 
+---+
|
|
|

Player O, Enter Position (1-9): 8

X | X | 
+---+
| O | 
+---+
| O | 
|

Player X, Enter Position (1-9): 3

X | X | X 
+---+
| O | 
+---+
| O | 
|

Play_ X Wins!
>>>
```

Result

Thus, the Python program for the **Tic Tac Toe Game** was executed successfully, and the winner was determined based on the game rules.

